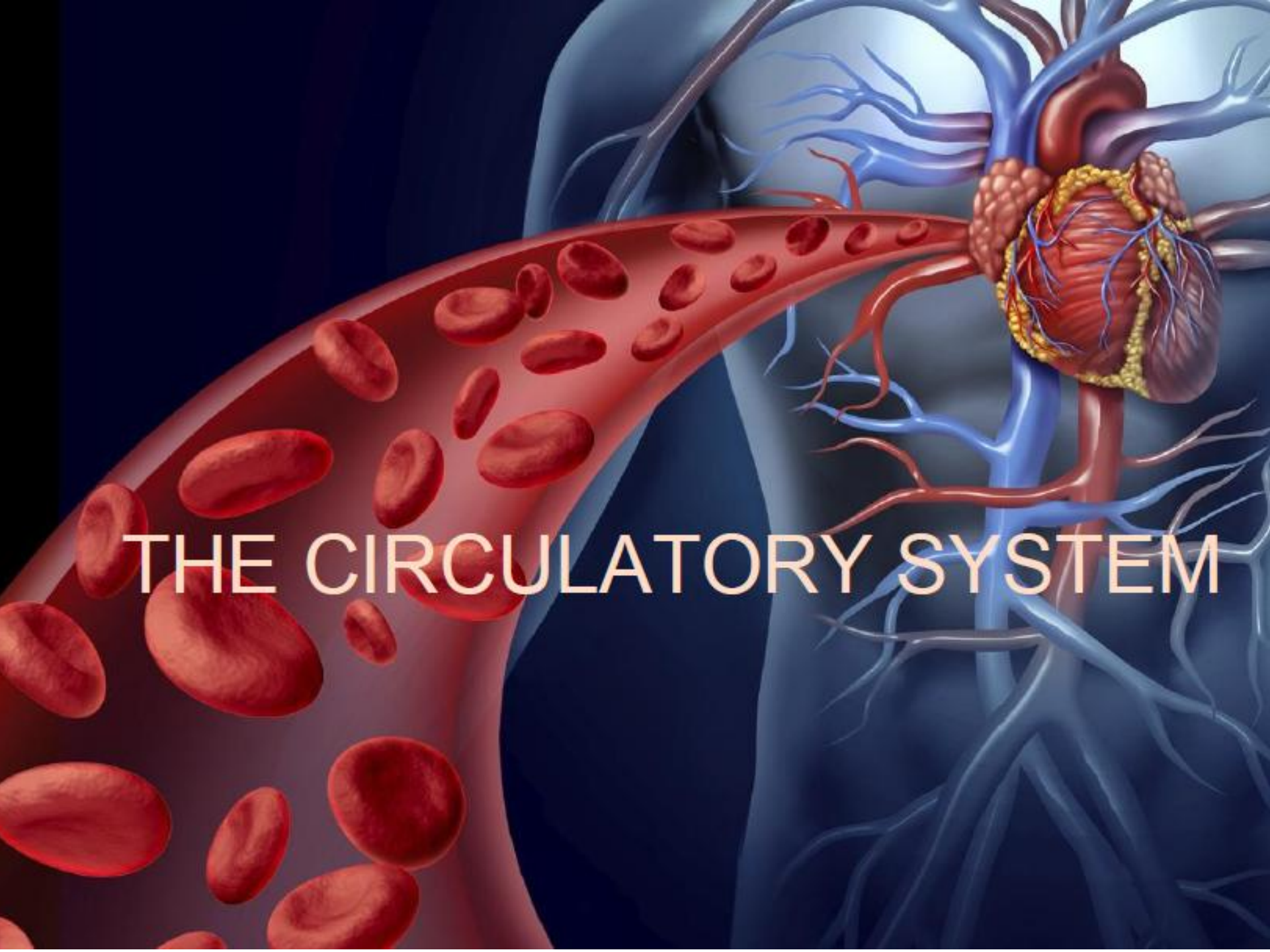
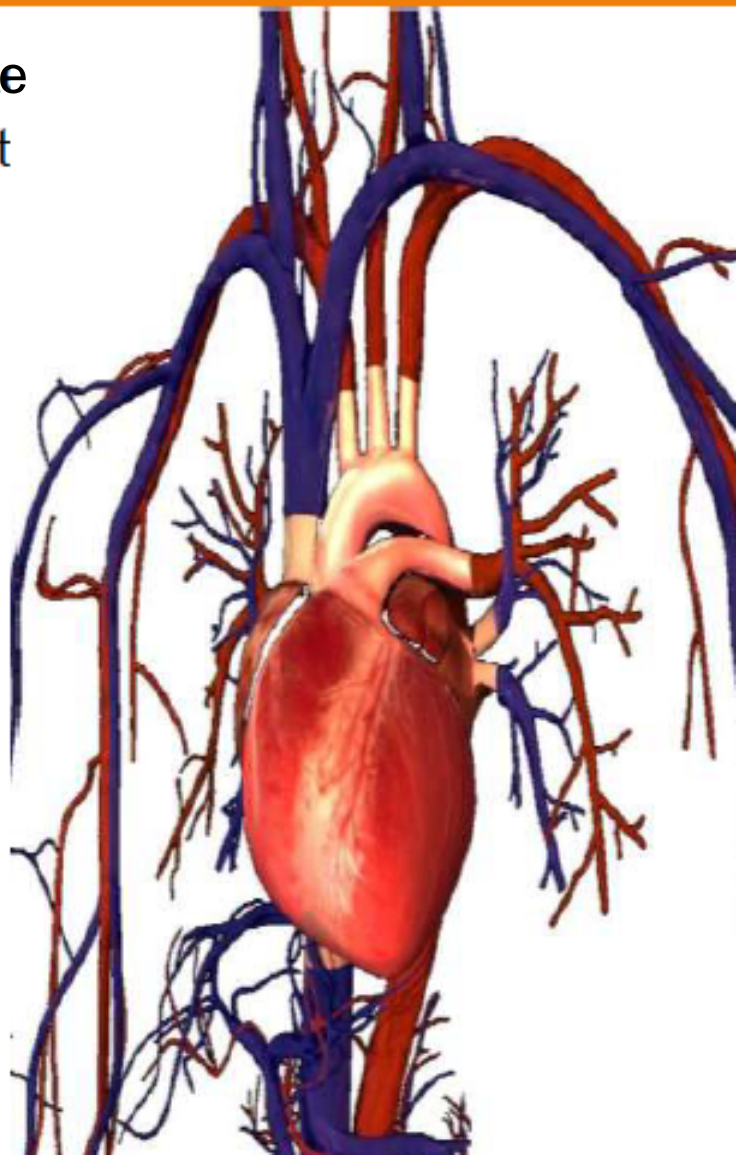

THE CIRCULATORY SYSTEM

Circulatory System

- The **Circulatory System** is responsible for transporting materials throughout the entire body.
- It **transports** nutrients, water, and oxygen to your billions of body cells and carries away wastes such as carbon dioxide that body cells produce.
- It is an amazing highway that travels through your entire body connecting all your body cells.



Functions of Human Blood Circulatory System

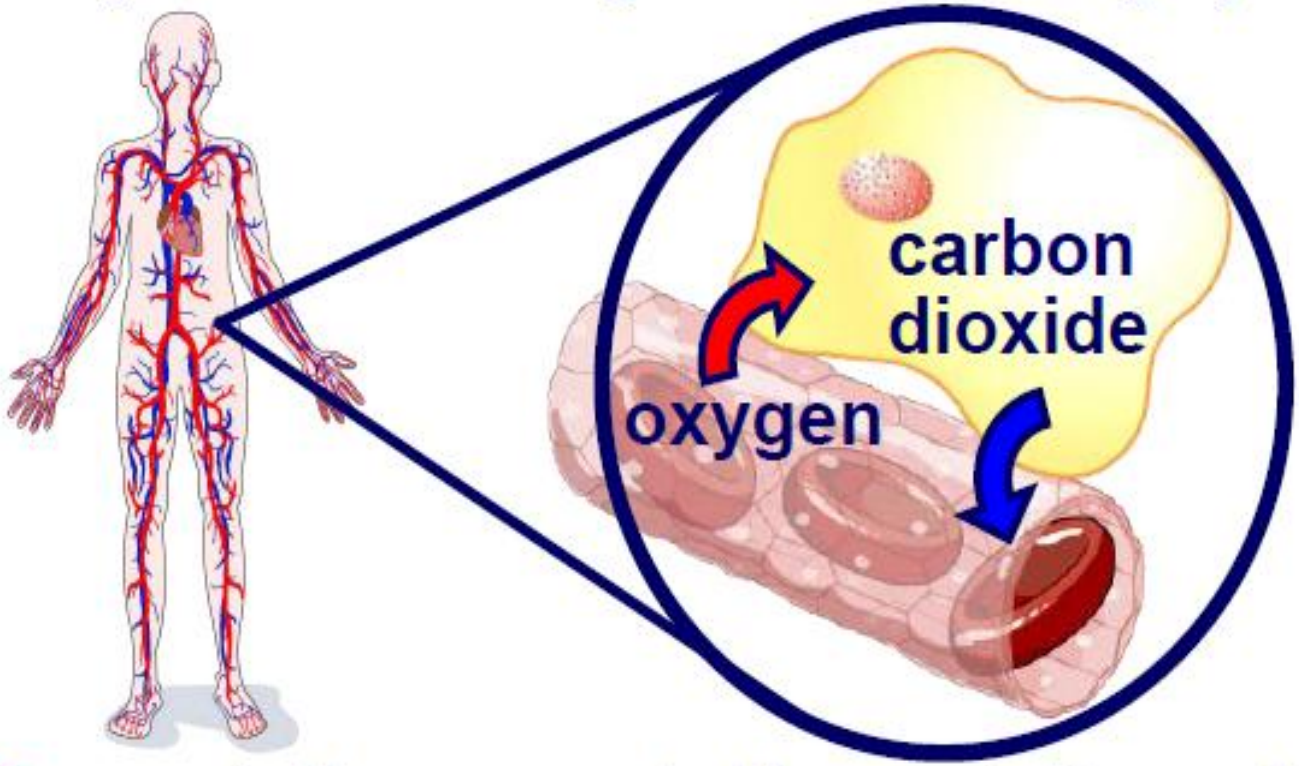
Includes transport of following

- 1. oxygen
- 2. carbon dioxide
- 3 nutrients
- 4. water
- 5. ions
- 6. hormones
- 7. antibodies
- 8. metabolic wastes



Circulatory System

- Which gases are transported to and from the body's cells by the blood flowing in the **circulatory system**?



Oxygen is the gas needed for respiration and is transported **to** the body's cells.

Circulatory System

- The circulatory system carries two types of blood

Oxygen-rich blood



- Blood travelling to the body cells
- High oxygen content
- Low carbon dioxide content

Oxygen-poor blood



- Blood travelling away from the body cells
- Low oxygen content
- High carbon dioxide content

- Arrangement of the circulatory system means that these two types of blood do not mix.

The human circulatory system is divided into two parts:

i) Blood vascular system
ii) Lymphatic system

i) Blood vascular System has following components:

- Blood
- Blood vessels- arteries, veins and capillaries
- Heart

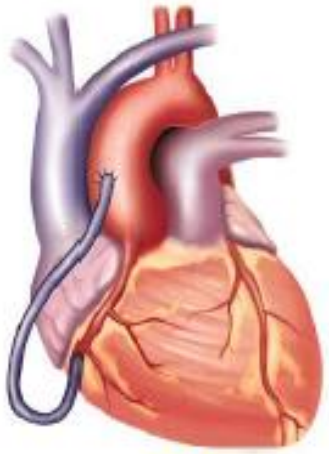
ii) Lymphatic System has following components:

- Lymph
- Lymph vessels

Circulatory System

Components

- Heart
- Blood
- Vessels
  - Arteries
  - Veins
  - Capillaries



Heart



Blood



Veins



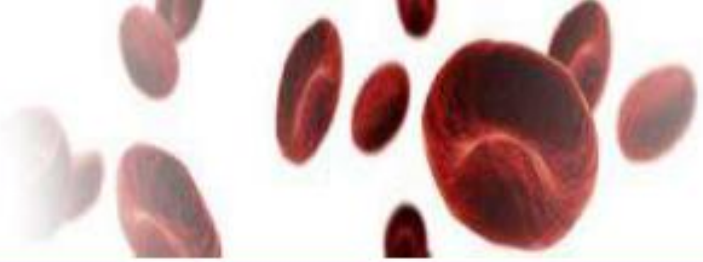
Arteries

Blood

- What percent of your body is blood? 8%
- How much blood do we contain?
  - On average 4-6 liters
  - We contain about a pint of blood for every 15 pounds of body weight
- Composition of Blood:
  - What percent of your blood is cellular? 45%
  - What percent of your blood is plasma? 55%



Blood



- Composed of plasma and blood cells
- Types of Cells are:
  - Red Blood Cells
  - White Blood Cells
  - Platelets



- Plasma
  - Straw colored
  - 90% water
  - 10% dissolved gases, salts, nutrients, enzymes, hormones, wastes, and proteins.

Blood composition

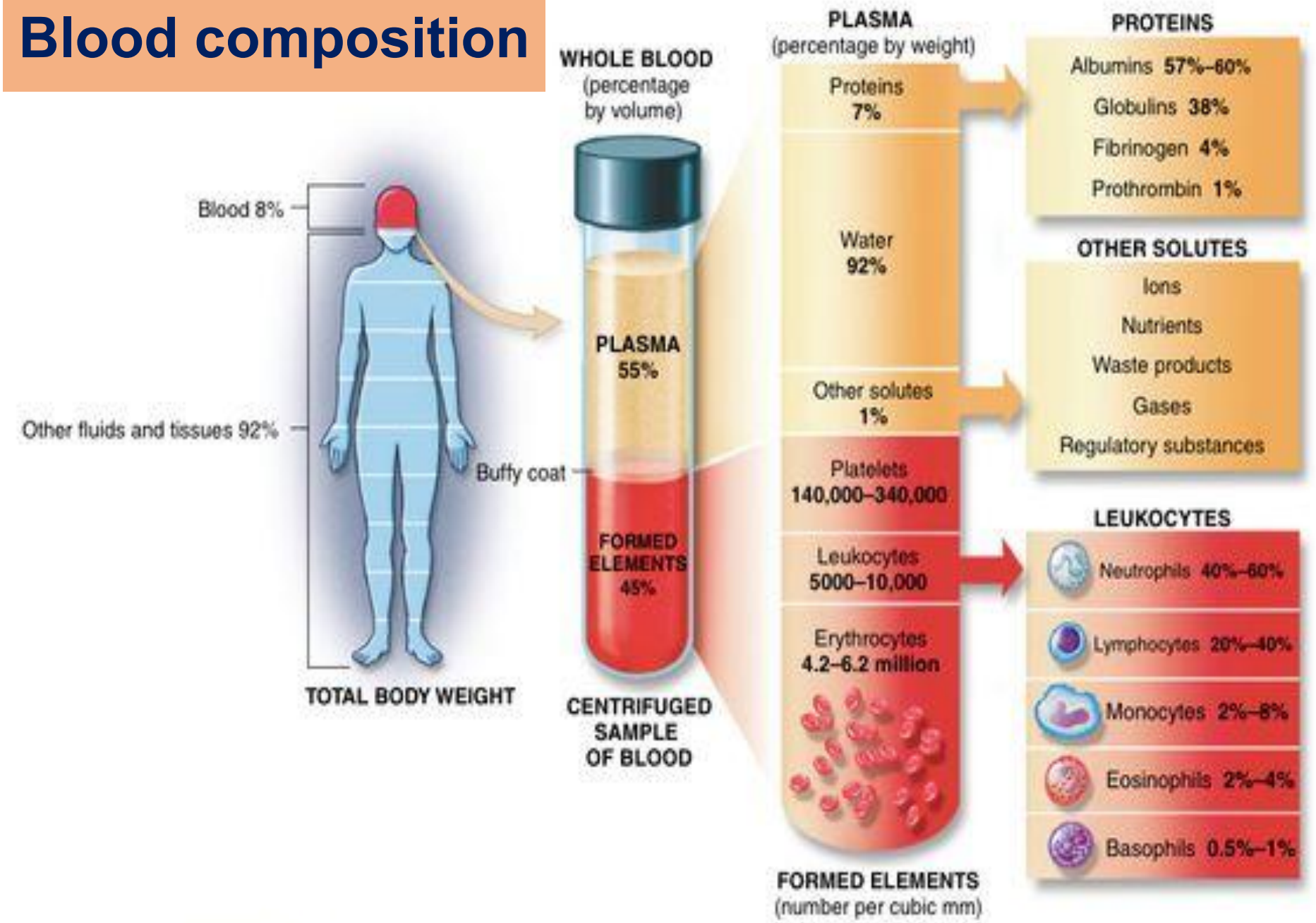
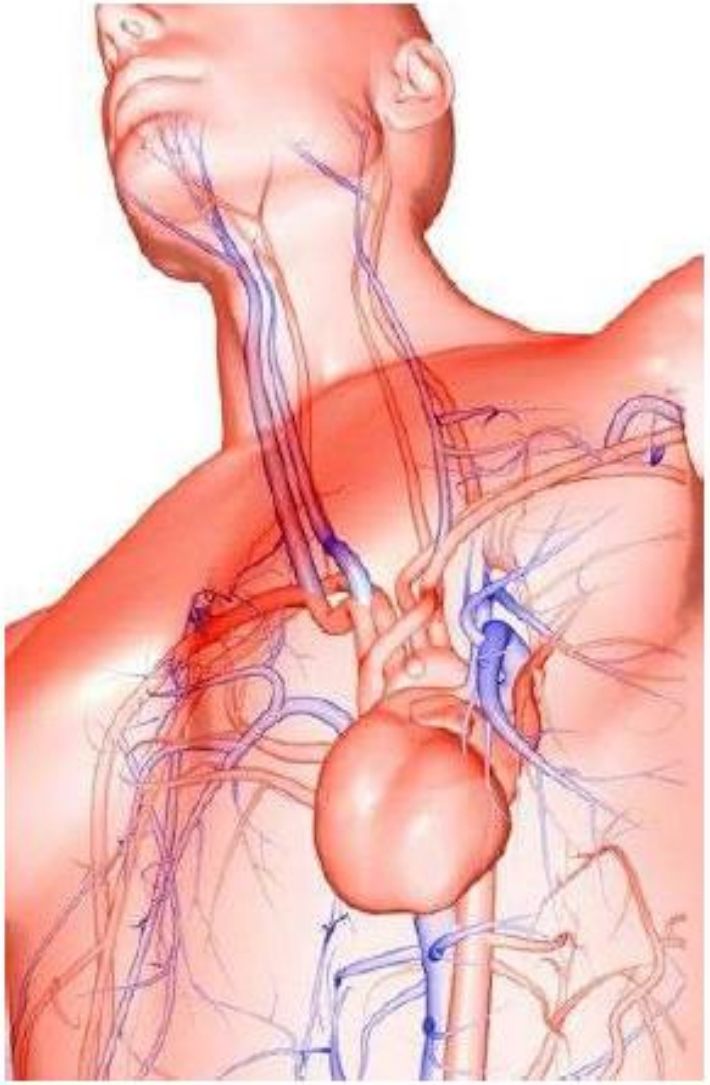
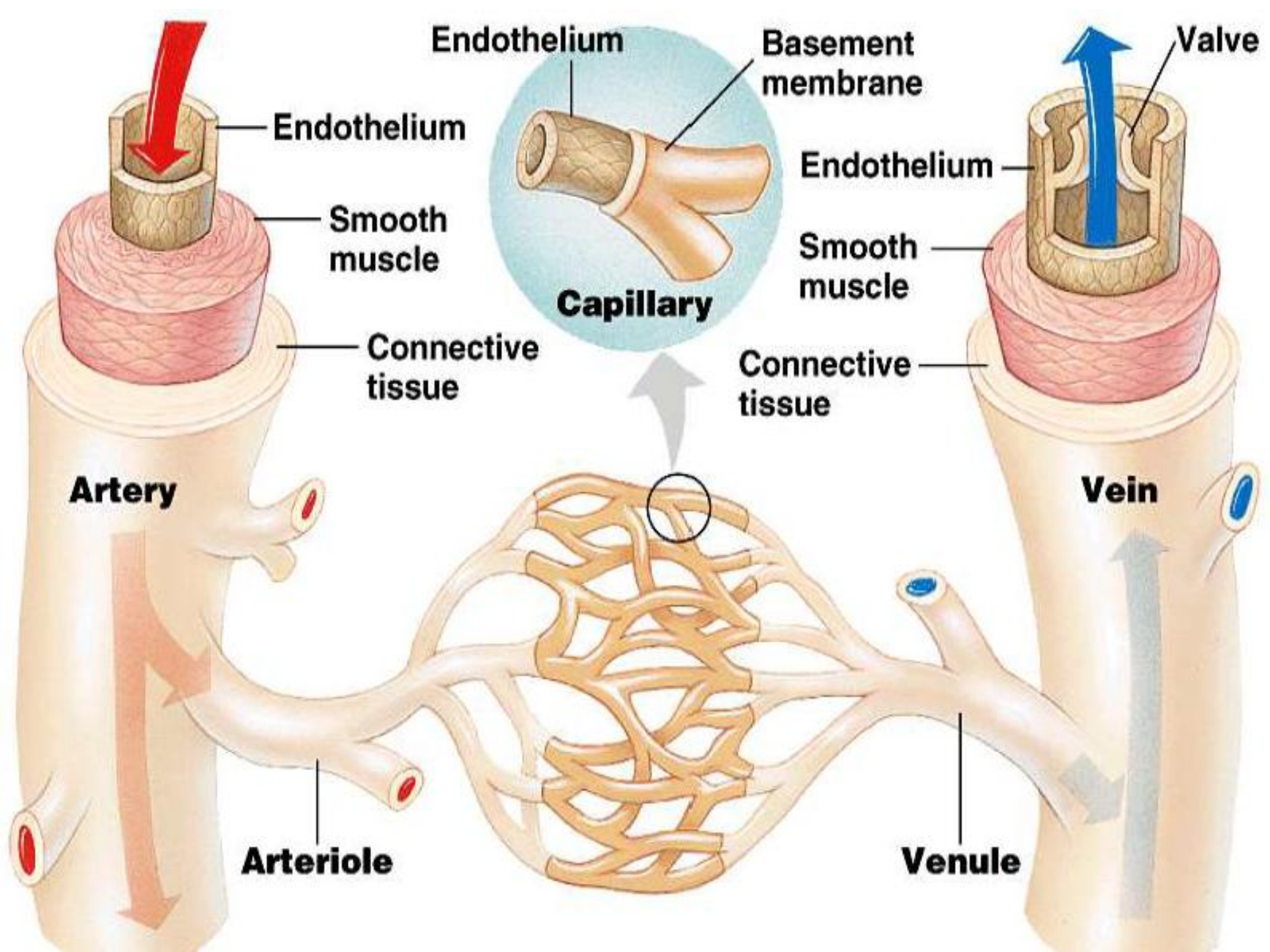


FIGURE 19-1 **Composition of Whole Blood.** Approximate values for the components of blood in a normal adult. (From Patton KT, Thibodeau GA: *Anatomy & Physiology*, ed 7, St Louis, 2010, Mosby.)

Blood Vessels

- As blood moves through the circulatory system it moves through 3 types of blood vessels:
  - **Arteries:** Carry blood away from the heart .
  - **Capillaries:** Link arterioles to veins.
  - **Veins:** Carry blood towards the heart




Endothelium
Basement membrane
Valve
Endothelium
Endothelium
Smooth muscle
Smooth muscle
Capillary
Connective tissue
Connective tissue
Artery
Vein
Arteriole
Venule

Heart



❖ **Cone shaped, muscular organ**

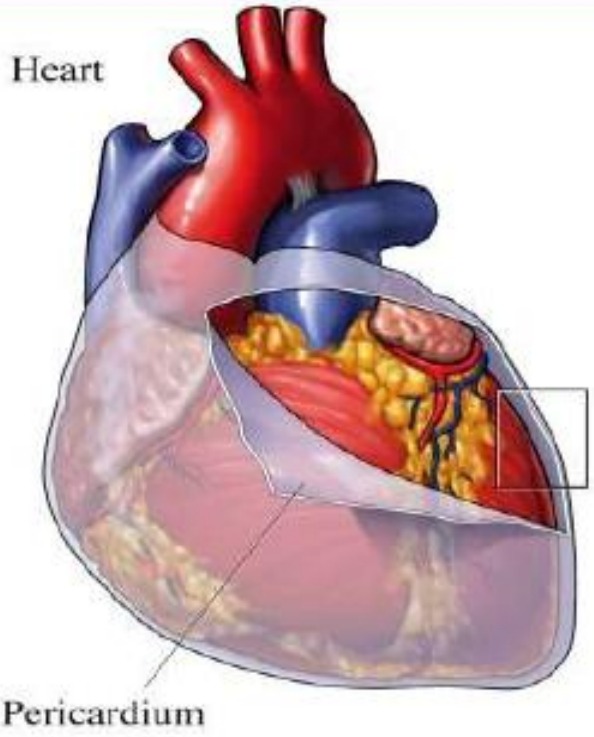
- The Heart Located near the center of the chest Hollow structure
- The Heart Enclosed in a protective sac called the ***pericardium***
- The heart contracts about ***72*** times per minute
  Pumps about **70ml** *of blood with each contraction.*

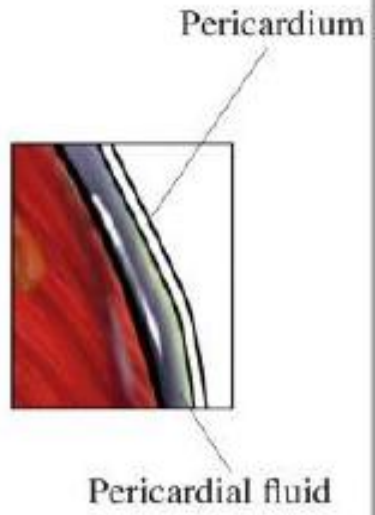


- The heart measures about 12 x 9 cm. and weighs about 300 gm. In males and 250 gm. In females .

The Heart: Structures

- Pericardium
  - Protective sac of connective tissue
  - Surrounds the heart
  - Filled with fluid





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The Heart



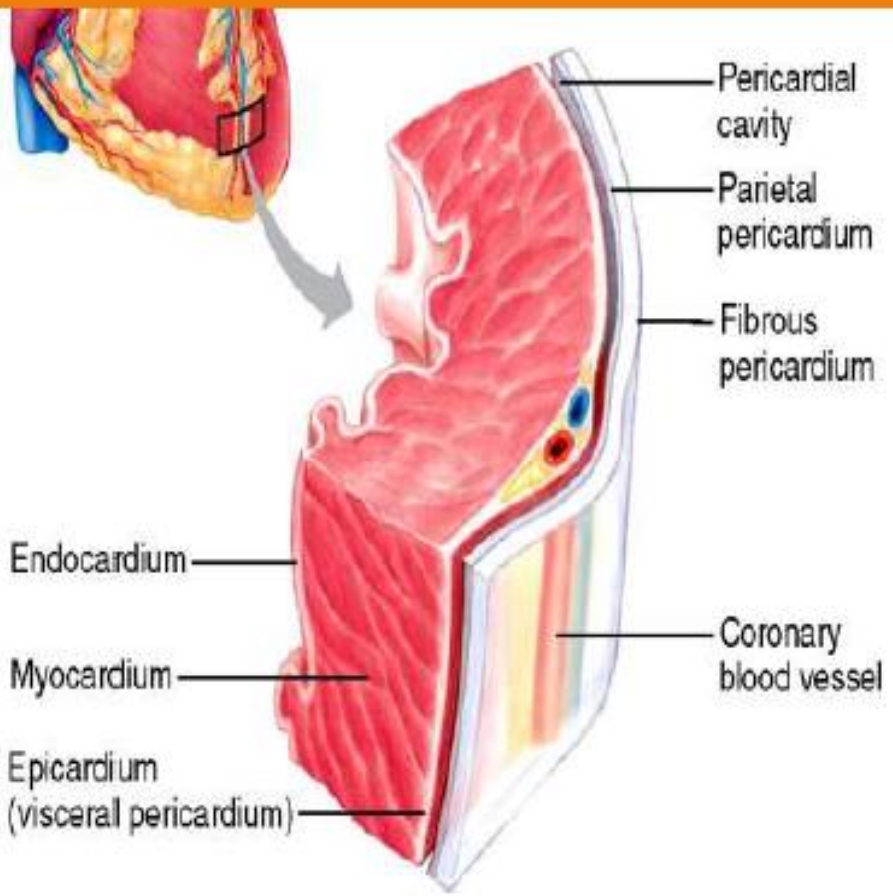
Structure

- Layers:
  - Epicardium **(Pericardium)**
  - Myocardium(Contractions of the myocardium pump blood through the circulatory system)
  - endocardium



The Heart: Structures

- **Myocardium**
  - The muscle of the heart
  - Strong and thick
  - Composed of spontaneously contracting cardiac muscle fibers
  - Can conduct electricity like nerves.
  - It's blood supply comes from the coronary arteries.

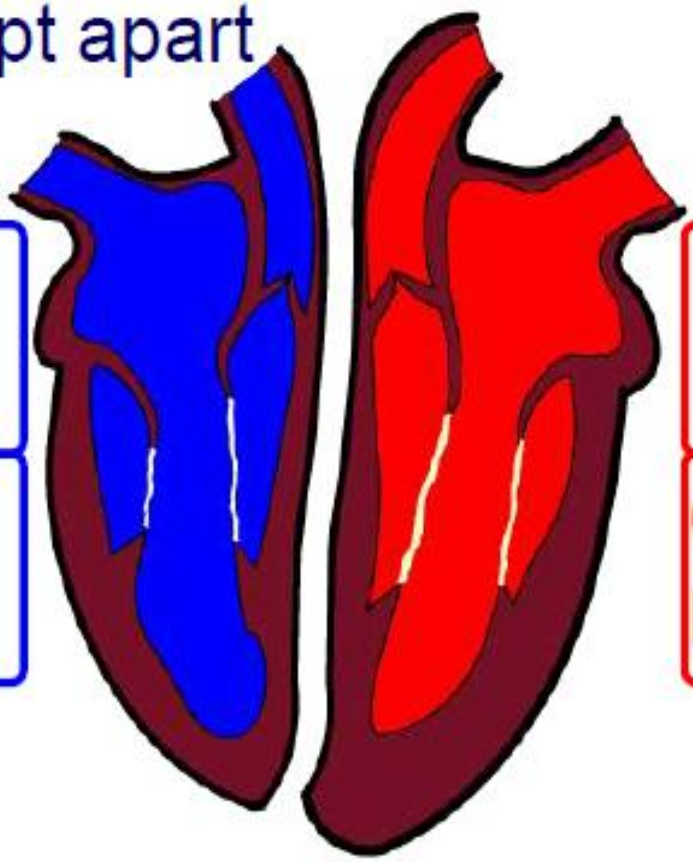


Circulatory System

- The inside of the heart is **divided into two sections** so that the **two types of blood** (**oxygen-rich** and **oxygen-poor**) are kept apart

Right side of the heart

Oxygen-poor blood

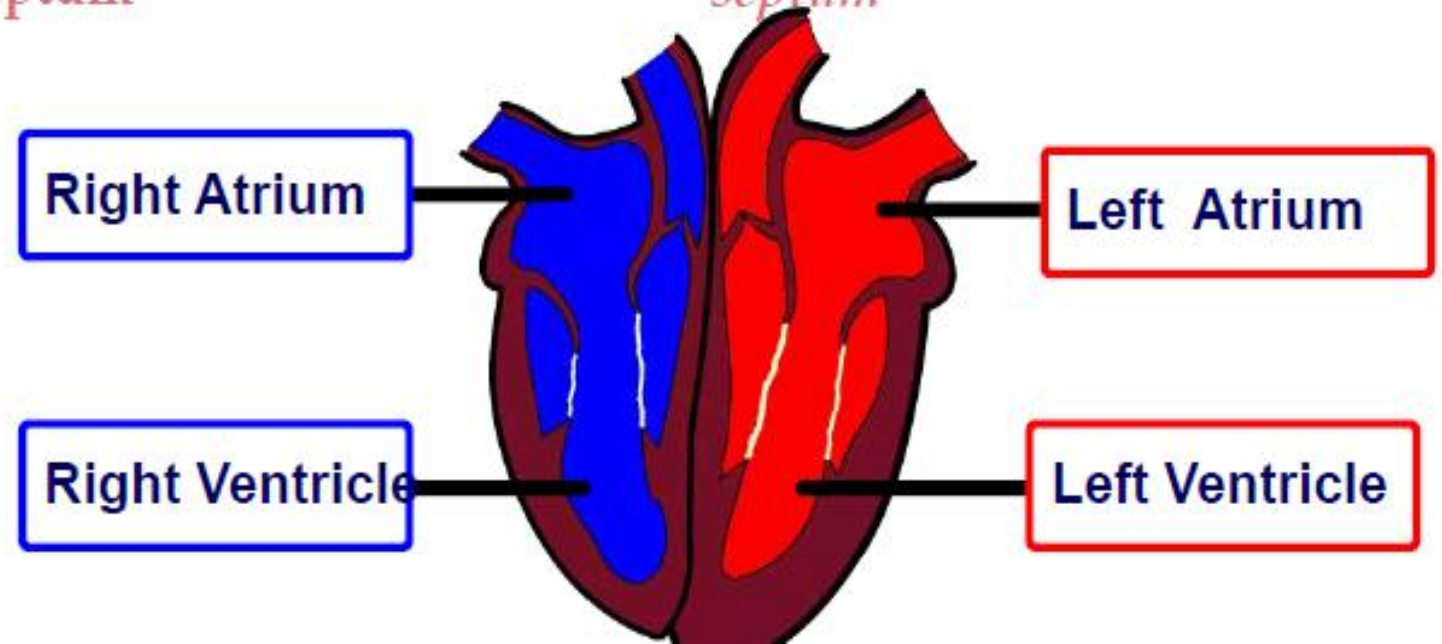


Left side of the heart

Oxygen-rich blood

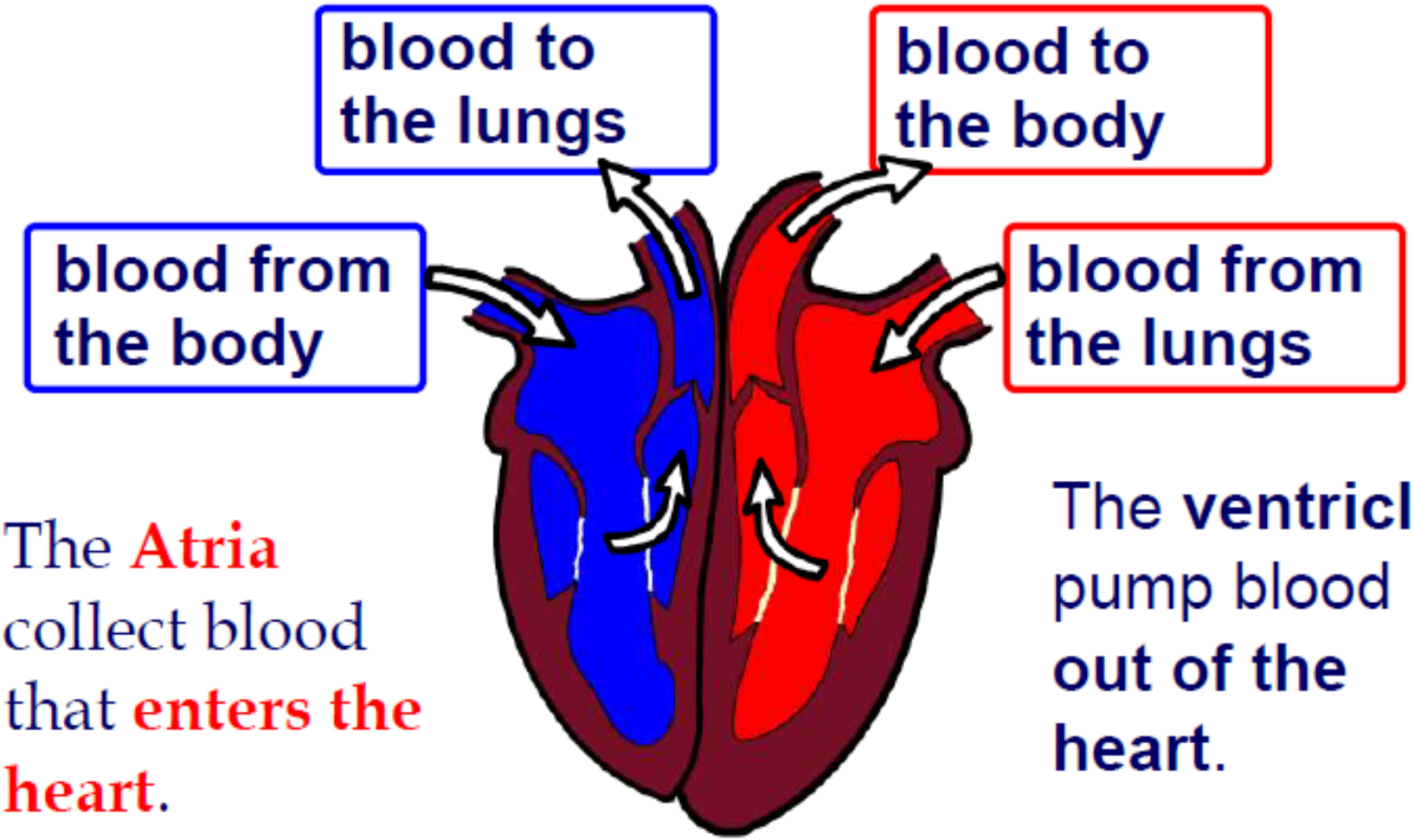
The Heart: Structures

- Four chambers
  - Two *Atria*
    - Upper chambers
    - Left and right
    - Separated by interatrial septum
  - Two Ventricles
    - Lower chambers
    - Left and right
    - Separated by *interventricular septum*



The Heart: Structures

- The chambers of the heart have different functions:



The Atria collect blood that **enters the heart**.

The **ventricles** pump blood **out of the heart**.

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The Heart



- Chambers:
  - Two atria (right and left)
  - Two ventricles (right and left)
- The right and left sides of the heart are separated by a septum, or wall. The septum prevents the mixing of oxygen rich and oxygen poor blood.
- The upper chamber (receives blood) is the atrium. The lower chamber (pumps blood out of heart) is the ventricle



Septum- separation by walls between chambers

Interatrial septum : Walls between two atrium, left and right

Intraventricular septum : Walls between two ventricles, left and right

Atrio-ventricular septum : walls between atrium and ventricles.

- The right and left sides of the heart are separated by a **septum**, or wall.
- The septum prevents the mixing of oxygen rich and oxygen poor blood.
- On each side of the septum are two chambers.
- The upper chamber (receives blood) is the atrium.
- The lower chamber (pumps blood out of heart) is the ventricle.

The Heart

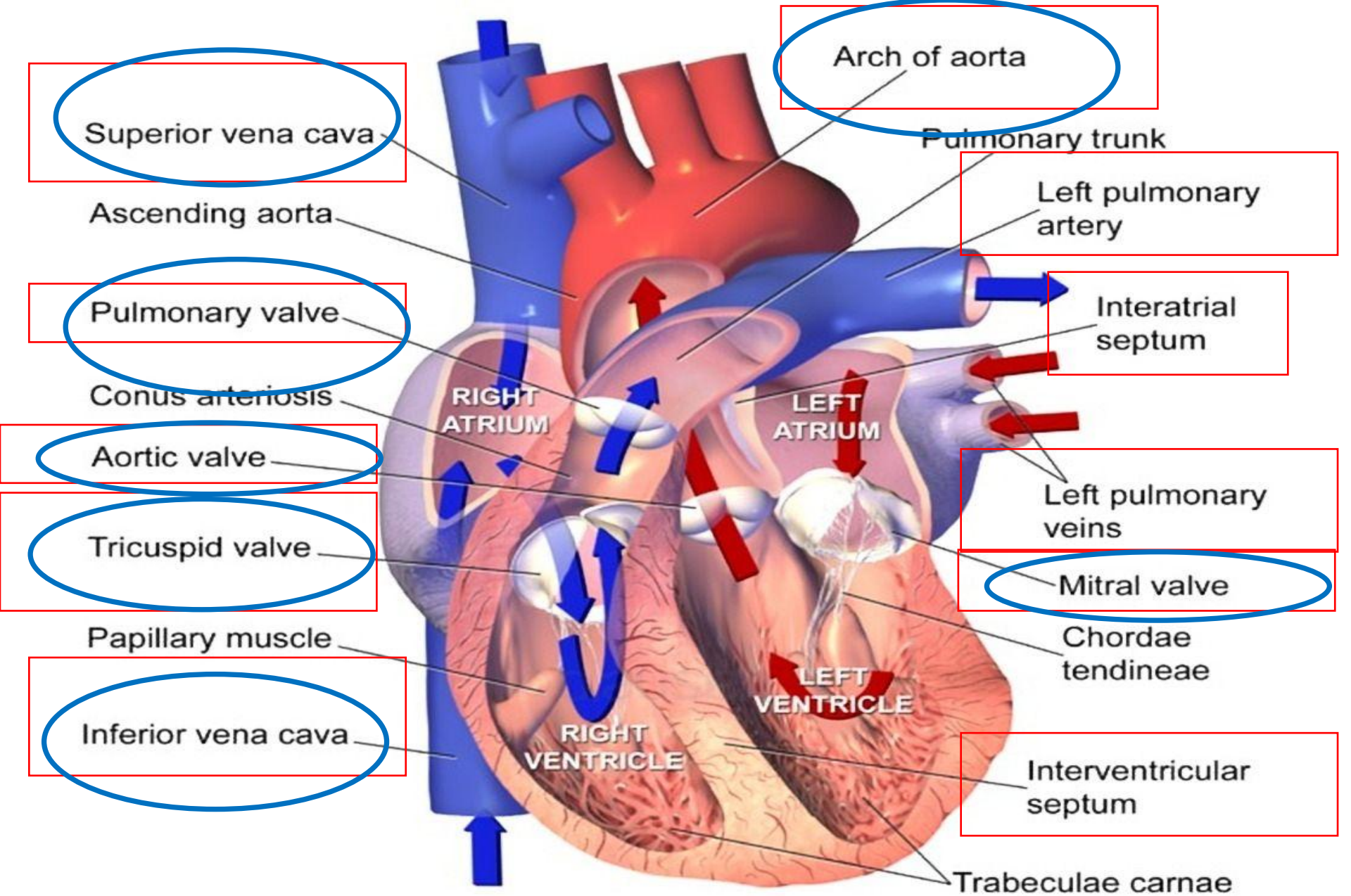


Valves

- Two atrioventricular valves:
  - tricuspid and bicuspid (mitral)
- Two semilunar valves:
  - pulmonary and aortic

- Flaps of connective tissue, called valves, are located between the atria and ventricles.
- Blood moving keeps the valves open. When the ventricles contract, the valves close which prevent blood from flowing back into the atria.
- Each atrium and ventricle contracts as a unit.





Sectional Anatomy of the Heart

Overview of Human circulatory system

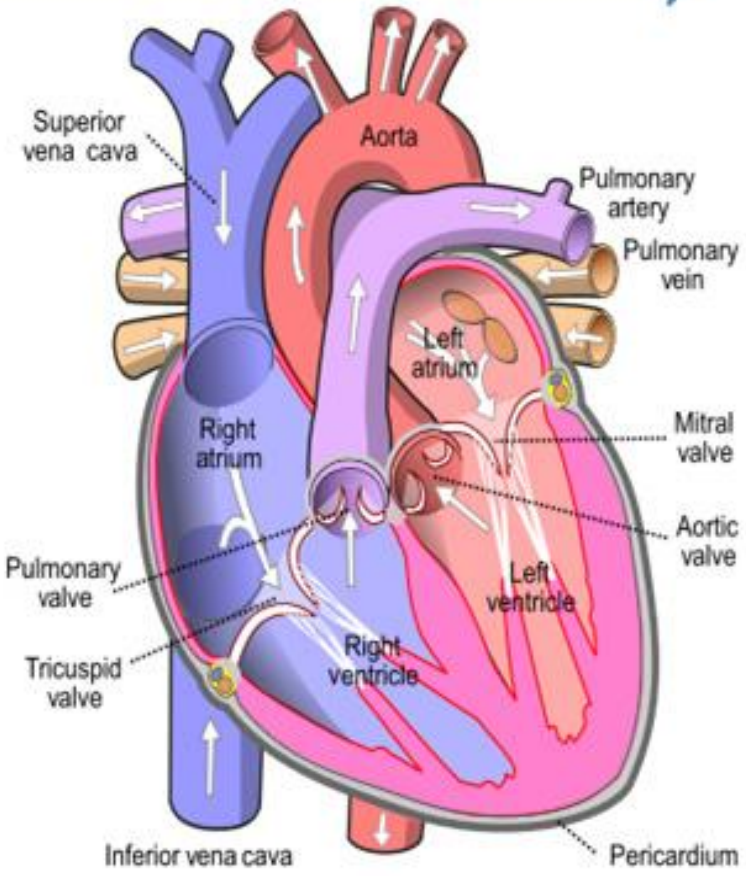
- **Function of circulatory system**: The overall function of the circulatory system is to transport metabolites and gases, remove waste products generated in the body, and transport of hormones to target tissues.

- **Closed type** Blood flows around the body through the specific blood vessels. Aorta, arteries, arterioles, venules, veins, and vena cava are specific blood vessel found in human circulatory system. The path of flow of blood in a closed system is given below:

Blood from the heart (oxygenated and pure)→ aorta→ arteries → arterioles→ capillaries → tissues → capillaries → venules → veins → vena cava → back to heart (deoxygenated and impure).

Path of blood flow

Deoxygenated blood



Right atrium → 2 large veins supply

Superior vena cava carries **Deoxygenated blood** from head, neck &upper limb.

Inferior vena cava carries **Deoxygenated blood** from lower parts of body.

Deoxygenated blood from right atrium enters the **right ventricle** through **Tricuspid valve**

Deoxygenated blood from the right ventricle enters **lungs** through **pulmonary artery.**

Oxygenated blood from **lungs** is supplied to the **left atrium**

Oxygenated blood from **left atrium** enters **left ventricle** through **mitral/bicuspid valve.**

Arch of Aorta arises from **left ventricle** which pumps **oxygenated blood** to **all parts of the body**

What is Double circulation and its types?

A mechanism in which blood circulates twice through the heart in one complete cycle is known as double circulation.

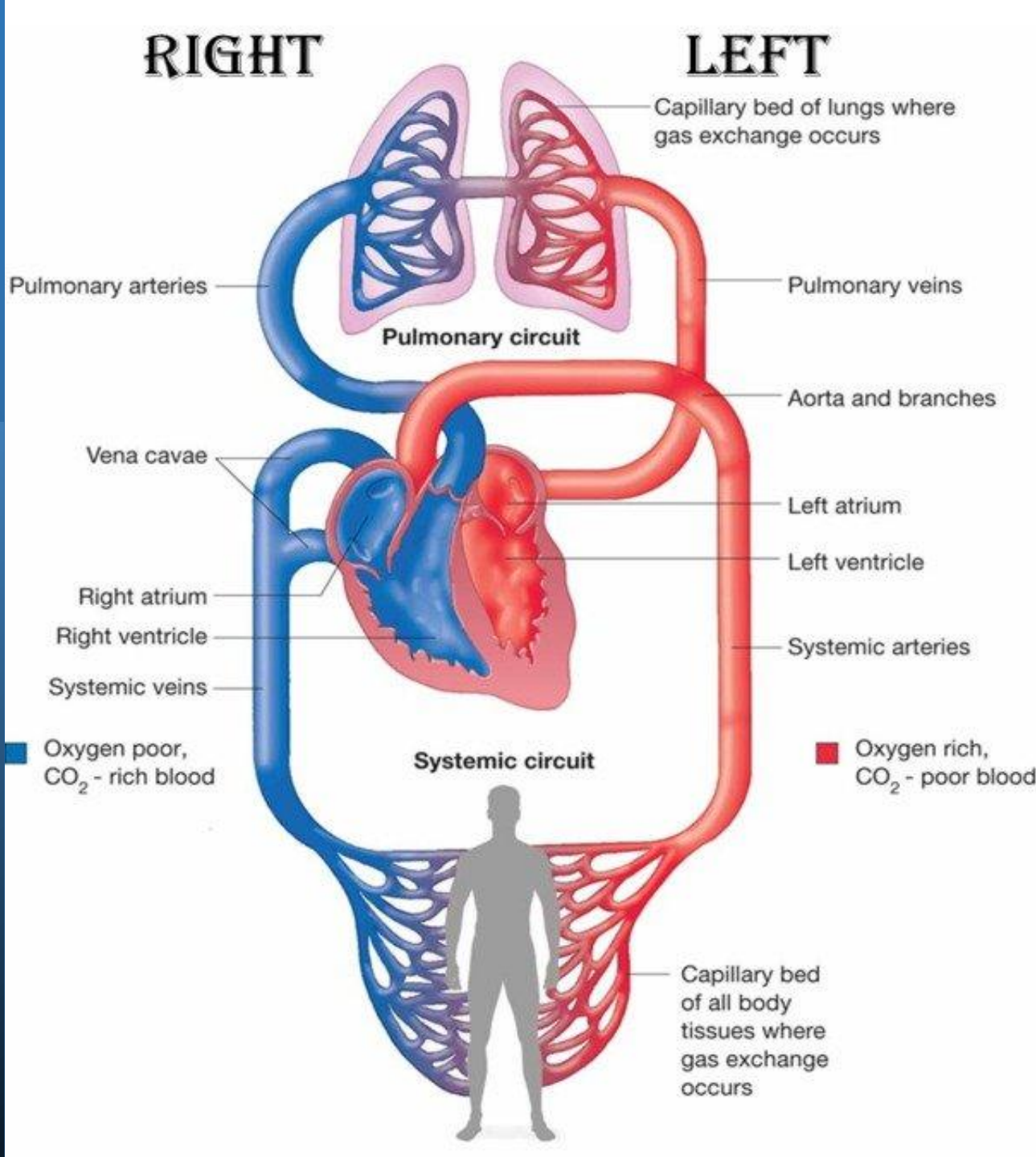
1. Pulmonary (lungs) Circulation

- Deoxygenated blood or impure blood from right ventricle of the heart carried to lungs and oxygenated blood back to the left ventricle of the heart (heart & lungs)

2. Systemic (body) Circulation

❖ Includes all arteries and arterioles that carry Oxygenated blood from the left ventricle to systemic capillaries, plus the veins and venules that carry deoxygenated blood returning to the right atrium after flowing through body organs. It has three subdivisions: **Coronary circulation, cerebral circulation and hepatic portal circulation.**

Double circulation: Pulmonary and Systemic



Blood vessels of the heart for pulmonary circulation

Pulmonary artery- Deoxygenated blood travels to the lungs (left as well as right). It is the only artery that carries deoxygenated blood for purification in the lungs.

Pulmonary veins- Oxygenated blood returns to the heart (left side from the lungs). The pulmonary veins are the only veins that carry oxygenated blood.

Aorta: large artery that carries oxygenated blood to small arteries and distribute them to different parts of body.

Superior Vena Cava- large veins that returns deoxygenated blood from head and arms to the heart.

Inferior Vena cava-large veins that returns deoxygenated blood from abdominal organs and legs to the heart. Blood vessels from abdominal organs such as gastrointestinal tract first enters the hepatic portal circulation and then joins the inferior vena cava.

Blood vessels involved in systemic circulation

1. Arteries:

- Carries oxygenated blood away from the heart to different body parts.
- Arteries are large, elastic, thick-walled and muscular. They expand to fill with blood and then propel it forward as they return to their normal diameter.
- Blood is carried under high pressure.
- Arteries ends up in arterioles in tissues.

2. Veins

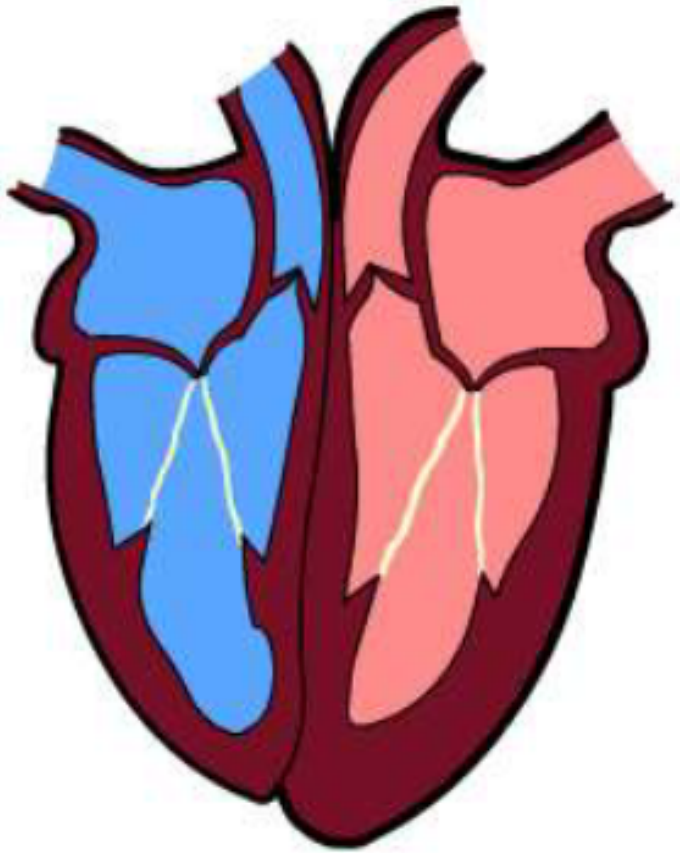
- Carries deoxygenated blood or impure blood from different body parts towards the heart.
- Veins are less muscular.
- Valve is present which is called vein valve. The presence of valve prevent blood flowing backwards under the force of gravity.
- Venules smaller channel take blood to the veins.

3. Capillaries

- Smallest vessels, microscopic, nutrients and gases diffuse here.

How does the heart pump blood?

- All the parts of the heart on either side, work together in a repeated sequence.
- The two atria contract and relax; then the two ventricles contract and relax.
- This is how blood moves through the heart and is pumped to the lungs and the body.
- One complete sequence of contraction and relaxation is called a **heartbeat**.



Conduction system of the Heart

- **The conduction system of heart comprises of cardiac muscles.**
  - About **1 percent** of the cardiac muscle fibres are different from others because they can generate action potentials over and over and do so in a rhythmical pattern.
  - Continue to stimulate a heart to beat even after it is removed from the body

For example; during transplantation into another person—all its nerves have been cut but still heart beating is there.

- The nerves (nervous system) regulate the heart rate but do not determine it.

The cells of cardiac muscle fibres have two important functions:

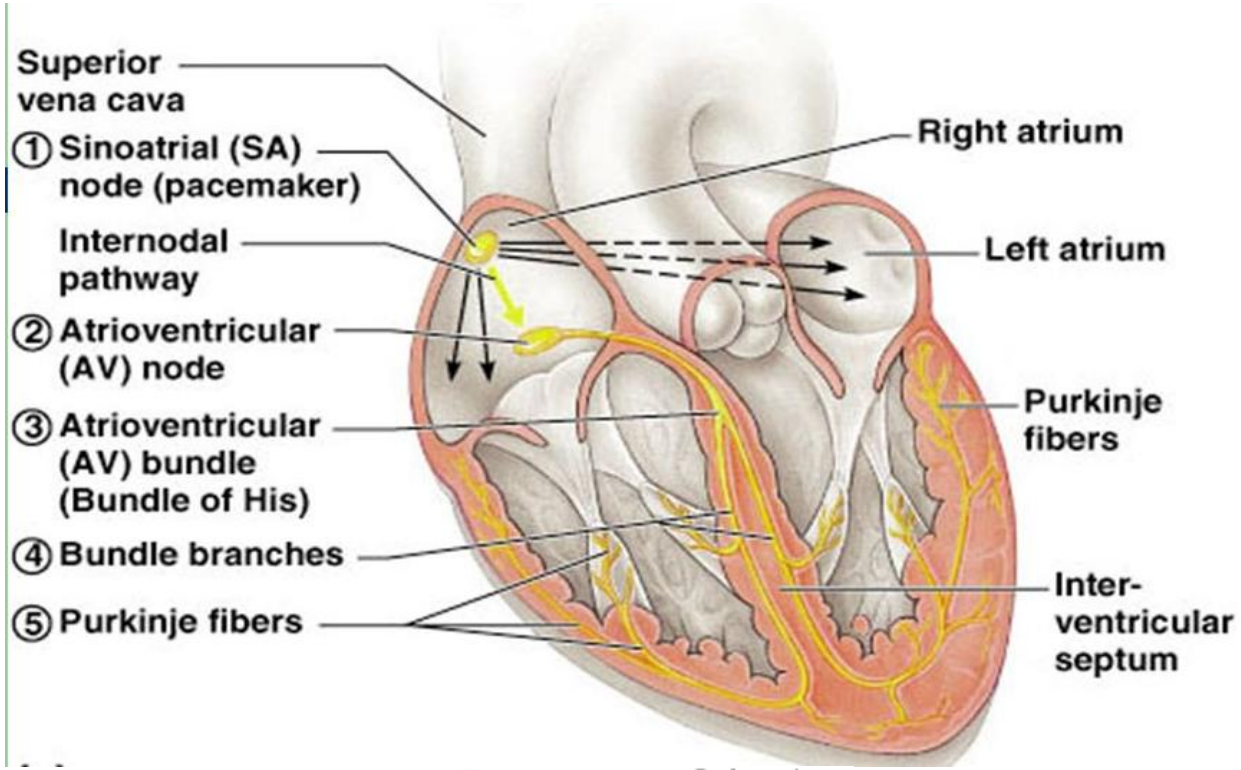
- **Act as pacemaker, setting the rhythm for the entire heart**, and
- **Form the conduction system, the route for action potentials throughout the heart muscle.**

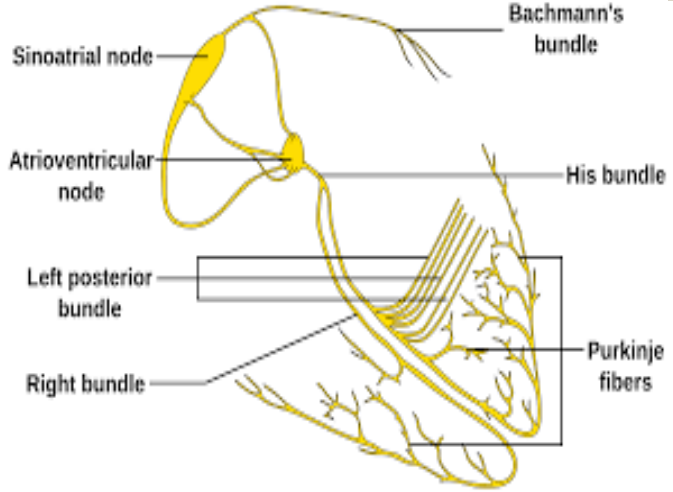
- **The conduction system ensures that cardiac chambers are stimulated to contract** in a coordinated manner, which makes the heart an effective pump
- Cardiac action potentials pass through the different components of the conduction system

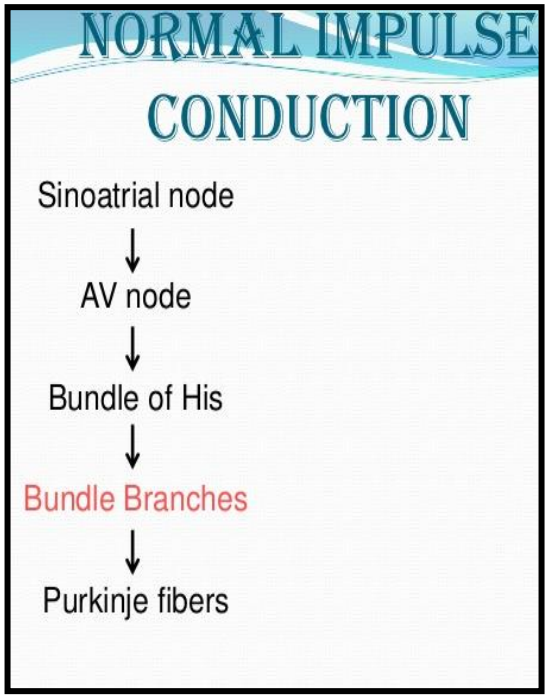
Properties of Cardiac Muscle Fibers

1. **Autorhythmicity**: The ability to initiate a heart beat continuously and regularly without external stimulation
2. **Excitability**: The ability to respond to a stimulus of adequate strength and duration (i.e. threshold or more) by generating a propagated action potential
3. **Conductivity**: The ability to conduct excitation through the cardiac tissue
4. **Contractility**: The ability to contract in response to stimulation

Arrangement of cardiac muscles of the heart

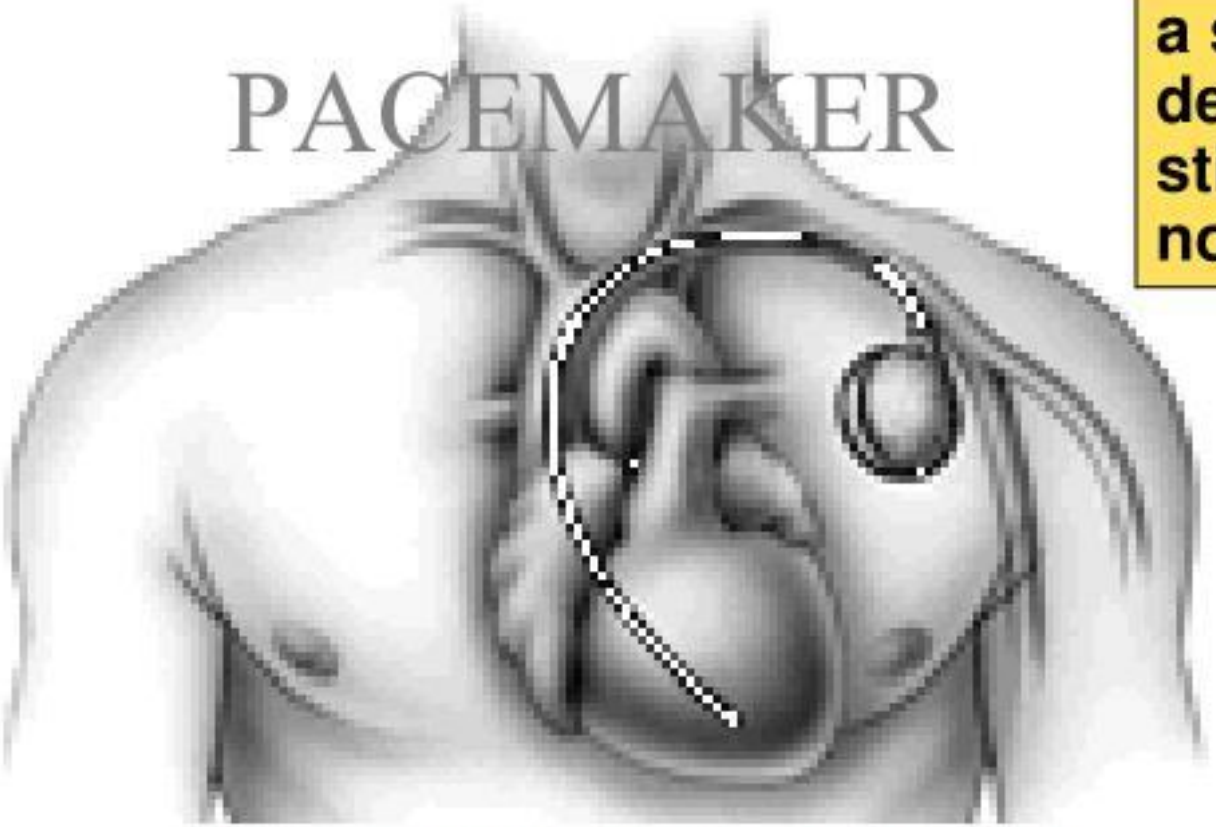






- **Sinoatrial node (SA node):** located in the right atrial wall just inferior to the opening of the superior vena cava. Cardiac excitation begins in the SA node. An action potential spontaneously arises in the SA node and then conducts throughout both atria via gap junctions of atrial fibres. Following, the action potential, the two atria finish contracting at the same time.
- **Atrioventricular node (AV node): AV node is in** the interatrial septum. The action potential reaches the AV node, where the action potential slows considerably, providing time for the atria to empty their blood into the ventricles.
- **Atrioventricular bundle (Bundle of His):** The action potential from the AV node, enters the Bundle of His. **The AV bundle is the only site where action potentials can conduct from the atria to ventricles.**
- **Right and left bundle branches :** After conducting along the AV bundle, the action potential then enters both the right and left bundle branches that course through the interventricular septum toward the apex of the heart.
- **Purkinje fibres:** These fibres rapidly conduct the action potential, first to the apex of the ventricles and then upward to the remainder of the ventricular myocardium. Thereafter, the atria contract, the ventricles contract within fraction of a second.

The SA node initiates the heartbeat and has been given the nickname of the "PACEMAKER"



a small electronic device that stimulates the SA node to fire

http://www.youtube.com/watch?v=CxJS0oEQeBQ

People with irregular heartbeats may have to have an artificial pacemaker 'inserted'.

Sinoatrial (SA) node is the pacemaker of heart

- The SA node sets the rhythm for contraction of the heart. It initiates action potentials about **100 times per minute faster** than any region of the conducting system.
- Various hormones and neurotransmitters can speed or slow pacing of the heart by SA node fibres. In a person at rest, for example, acetylcholine released by the parasympathetic division of the autonomic nervous system typically **slows SA node pacing to about 75 action potentials per minute, causing 75 heart beats per minute**.
- If the SA node becomes diseased or damaged, the slower AV node fibres can become pacemaker. With pacing by the AV node, however heart rate is slower, only **40 to 60 beats per minute**. If the activity of both nodes is suppressed, the heartbeat may still be maintained by the AV bundle, a bundle branch or Purkinje fibres. These fibres generate action potentials very slowly, about **20-35 times per minute**. At such a low heart rate, blood flow to the brain is inadequate.

Cardiac cycle

The sequence of event which occur from the beginning of one heartbeat to the beginning of the next heartbeat is called cardiac cycle.

The four events that occur in cardiac cycle are as follows :

a) **Atrial systole: An action potential from the SA node causes atrial depolarization. 70% of blood passively flows into the ventricles through the opened AV valves and rest 30 % is pumped into the ventricles by contraction of atria.**

b) **Ventricular filling and atrial diastole:** AV valve still open, and nearly 1/3rd of the ventricle is filled with blood at the end of atrial systole.

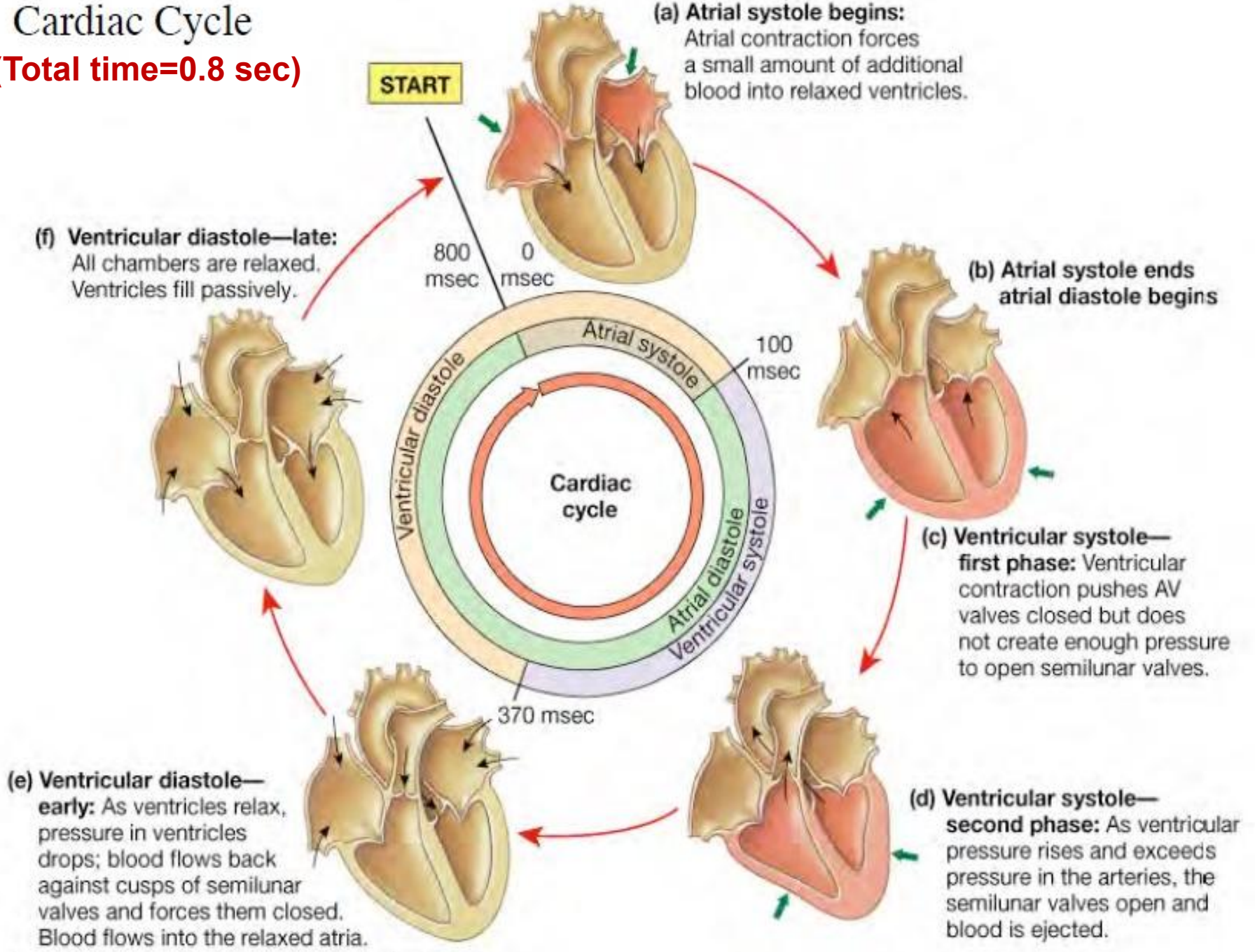
c) **Ventricular systole/ Ventricular depolarization.** When atrial systole comes to an end, the action potential from SA node passes to AV node as well as to the walls of ventricles. Consequently, ventricles contract which results in the rise of ventricular pressure and both the AV valves closes to prevent the back flow of blood. The contraction of ventricle is so strong that semilunar valves open. The blood is pumped into pulmonary artery and aorta from right and left ventricles, respectively. Ejection continues until the ventricles start to relax**. At rest, the volume of blood ejected from each ventricle during ventricular systole is about 70 ml.**

d) **Ventricular diastole**: Ventricular systole is followed by ventricular diastole during which there is fall in ventricular pressure. The semilunar valve closes. At the same time AV valve opens to allow the flow of blood from both the atria to respective ventricles.

The relaxation period begins at end of a cardiac cycle when the ventricles start to relax, and all the four chambers are in diastole.

Cardiac Cycle

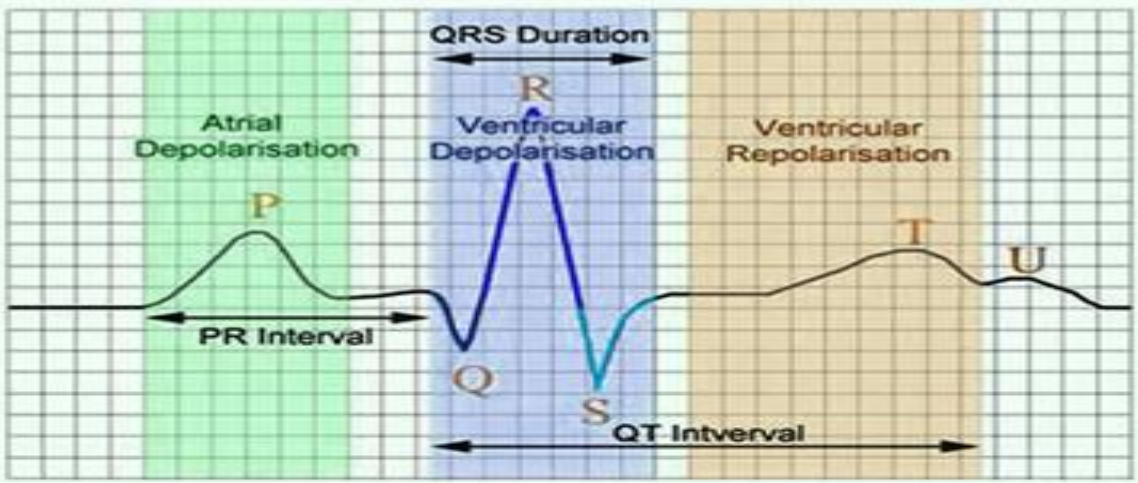
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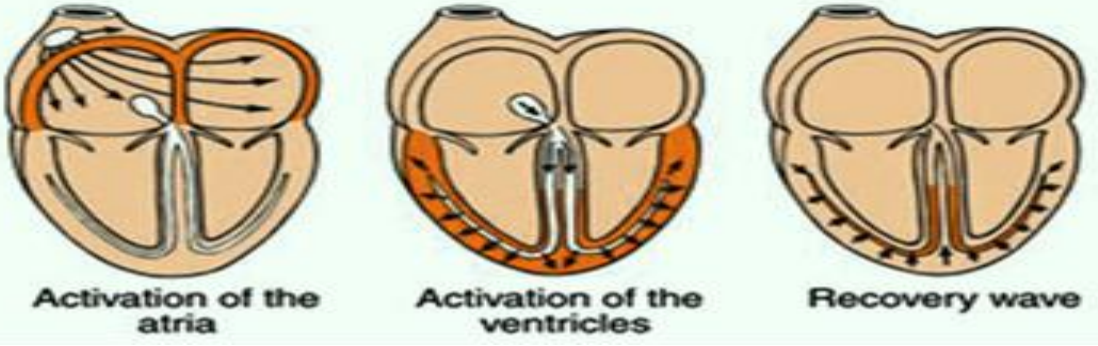


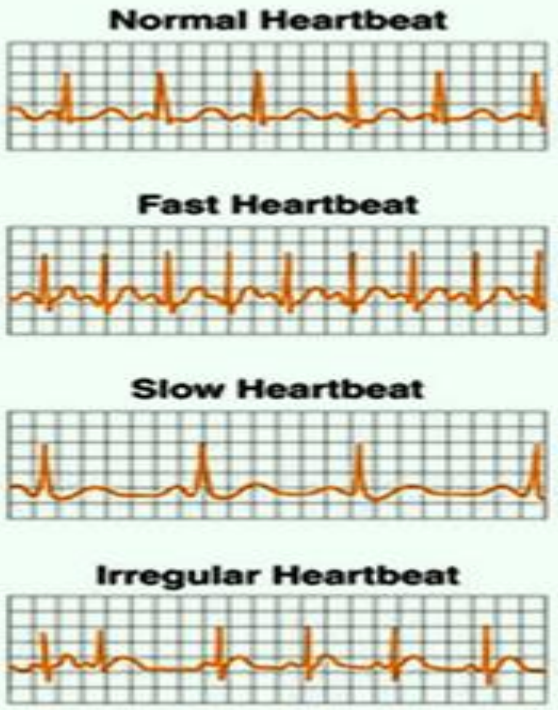
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Electrocardiogram (ECG)

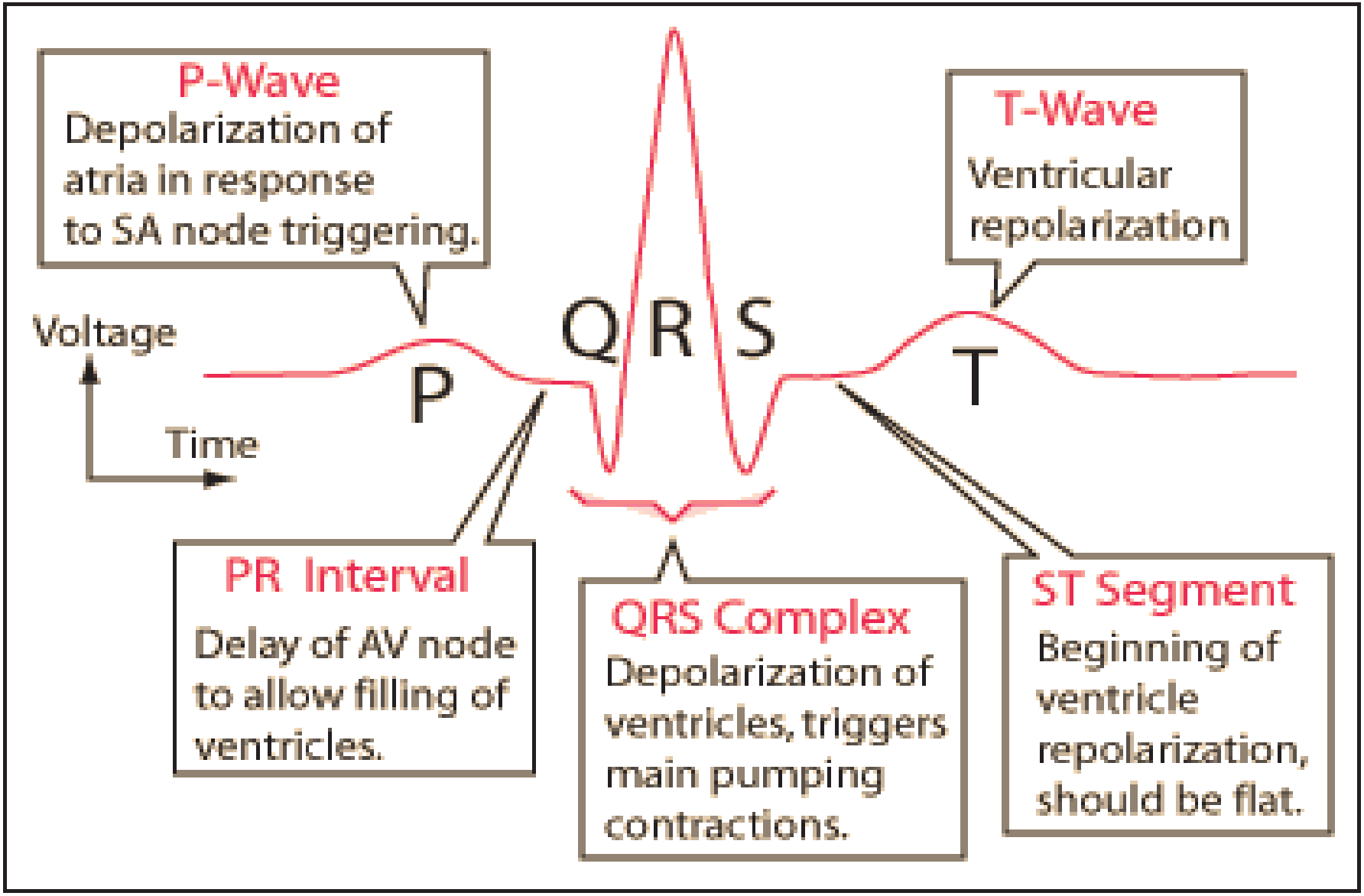
- Conduction of action potentials through the heart generates electrical currents that can be picked up by the electrodes placed on the skin.
- A recording of the electrical changes that accompany the heartbeat is called an electrocardiogram.
- Electrocardiogram is a graphical recording of electrical conduction in the heart over a period of time using electrodes placed over a skin.







Understanding the Graphics of normal (ECG)



Cardiac Output

- Defined as the amount of blood each ventricle pumps out per minute.
- Determined by:
  - **Stroke volume** – amount of blood that each ventricle pumps out per beat
  - **Heart rate** – number of times the heart beats in one minute
- **Cardiac Output = Heart rate X Stroke volume**

= 75 beats /minx 70 ml/beat

= 5250 ml/min or **5.25 litres /min**

- **Factors such as exercise increase stroke volume or heart rate and eventually increase cardiac output**

www.fppt.info

What is Heart Sound?

The rhythmic closure and opening of the heart valves cause the sound of heartbeat.

a) Closing of AV valve causes “lubb sound”.

b) Closing of semilunar valve at the end of ventricular systole causes “dubb sound”.

What is Heartbeat and pulse?

- The human heart beats at the rate of about 72-80 per minute in resting condition.
- **Arterial pulse:** It is a wave of distension that passes along the arteries immediately after a ventricular systole. The pulse rate increases during exercise, fever, and emotional and psychological excitements.

What is a Blood pressure?

- Blood pressure is a measure of the force exerted by the blood on the wall of the arteries.
- An example is 120/80 (systolic pressure/diastolic pressure).
- **Systolic pressure** is the result of the contraction of the ventricles (normal 110-140)
  Diastolic pressure is during the ventricle relaxation (normal 70-90)

DISORDERS OF CIRCULATORY SYSTEM

1. Hypertension (High Blood Pressure)

- When the blood pressure is higher than normal BP, it is called hypertension.
- If an individual repeatedly has the BP of 140/90 or above, it shows hypertension.
- It leads to heart diseases and affects vital organs (brain, kidney etc).

2. Coronary Artery Disease (CAD) or Atherosclerosis

- Here, Ca, fat, cholesterol and fibrous tissue are deposited in coronary arteries.
- It makes the lumen of arteries narrower and thereby affects the blood supply.

3. Angina (angina pectoris)

- An acute chest pain due to O_2 deficiency to heart muscles.
- It occurs due to improper blood flow.
- It is common among middle-aged and elderly.

4. Heart Failure (congestive heart failure)

- It is the inability of heart to pump blood enough to meet the needs of the body.
- Congestion of the lungs is the main symptom.

5. Cardiac arrest

Here, heart stops beating.

Cardiac Arrest is an electrical problem.

Heart Attack is a plumbing problem.

6. Heart attack

It is the sudden damage of heart muscle due to inadequate blood supply.

Cardiovascular (circulatory) system

1. Out of the following, which one is NOT the physical characteristics of the blood?

A. Volume is 15-16 liters in an average sized adult male
B. Slightly alkaline pH
C. Constitute 20% of extracellular fluid
D. Provide 8% of total body mass

2. Blood helps in transportation of the following components, except one, identify that?

A. Hormone
B. Oxygen
C. Heat
D. Lymph

3. Human has open blood circular system.

A. True
B. False

4. ECG records electrical changes in which of the following layers of the heart, mark the correct option.

A. Epicardium
B. Pericardium
C. Endocardium
D. Myocardium

5. The first heart sound is produced by the :

A. Closure of the aortic and pulmonary valves
B. Opening of the aortic and pulmonary valves
C. Closure of the mitral and tricuspid valves
D. Opening of the mitral and tricuspid valves

6. Normally, the rate of the heart beat in a human is determined by:

A. Bundle of His
B. all cardiac muscles
C. the SA node
D. the cervical ganglion

7. The ventricular repolarisation in ECG is best seen in:

A. "P" wave
B. "Q" wave
C. "R" wave
D. "T" wave

8. Stroke volume of each ventricle in normal adult is :

A. 30 ml
B. 130 ml
C. 70 ml
D. 5 liters

9. Cardiac output is a measure of:
A. Peripheral resistance * tissue fluid
B. Peripheral resistance * cardiac rate
C. Blood pressure * Cardiac rate
D. Heart rate * Stroke volume

10. Minimum blood Pressure is in:
A. Aorta
B. Arteries
C. Capillaries
D. Venules

11. SA node acts as heart's pacemaker
(a) because it has a poor cholinergic innervations
(b) because it has a rich sympathetic innervations
(c) because of its capability of generating impulses
(d) because it generates impulses at the highest rate

12. The ventricular muscles accepts impulses directly from
(a) AV node
(b) Bundle of His
(c) Right and left bundle branches
(d) Purkinje fibres